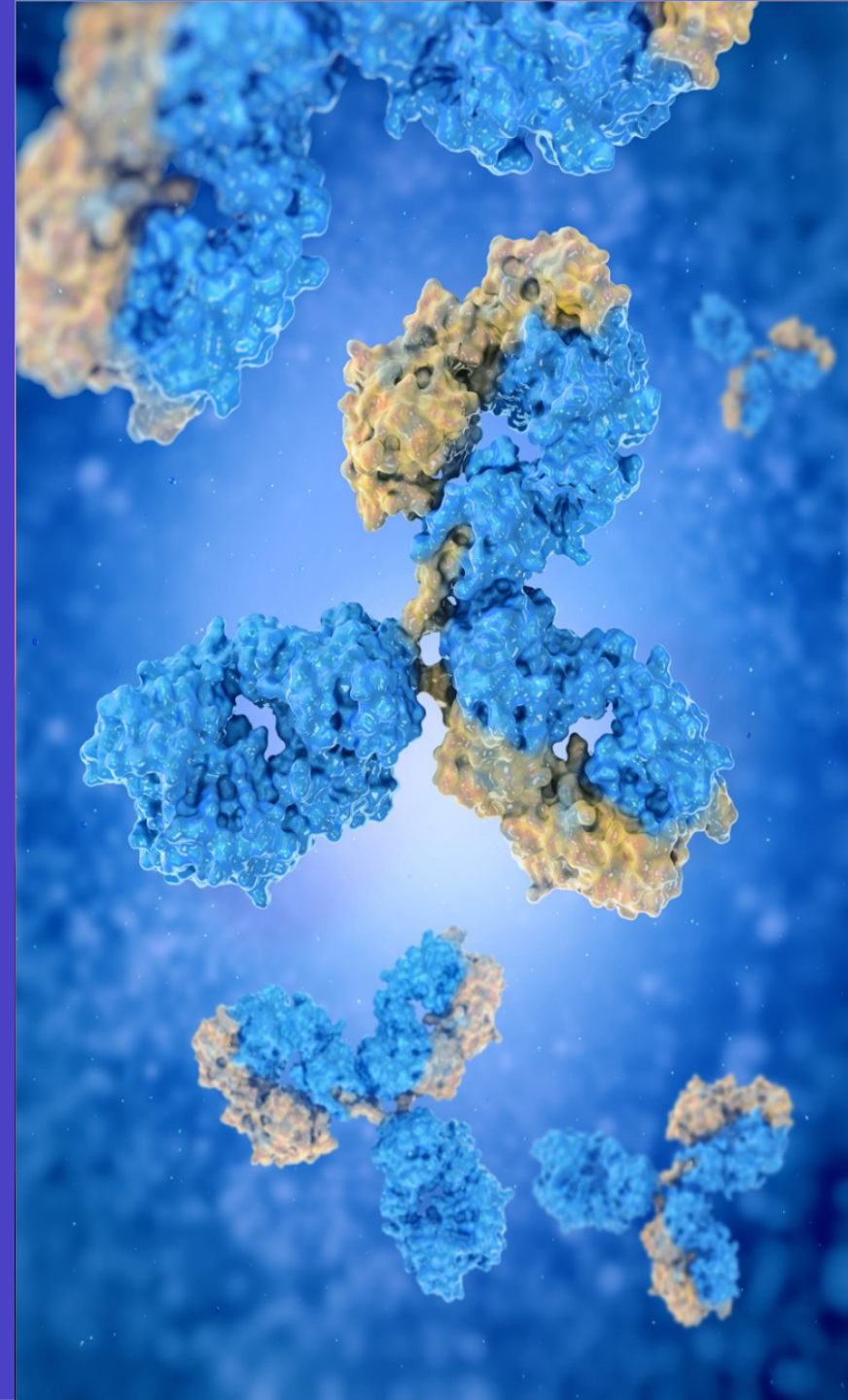


Downstream Processing of Antibodies

Andres Rodriguez



Introduction

What are antibodies?

- Y-shaped proteins **produced by immune system** → **defend against viruses & bacteria**
- Work by recognizing & binding to specific molecules (antigens) on pathogen's surface → **helps neutralize them & easier to destroy**
- **Highly specific** → used in vaccination to protect against future infections
- Designed from mammalian cell cultures

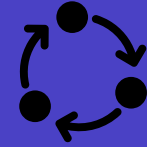
What is downstream processing?

- **Purification & separation of antibodies** from mixture of proteins & contaminants produced by biological system
- Includes several stages → clarification, initial, intermediate purification, and polishing (via chromatography, filtration, and centrifugation)
- *Goal:* isolate & purify antibody **while maintaining structural & functional integrity** for therapeutic use & diagnostic assays

How Downstream Processing Works

- 1. Harvest & clarification:** Antibody-containing cell culture medium → harvested & clarified to **remove cells, cell debris, large impurities**
- 2. Initial purification:** Clarified medium subjected to protein A chromatography allowing for **separation from other proteins**
- 3. Intermediate purification:** Additional chromatography steps → ion exchange & size exclusion chromatography **further purify & separate antibody from other impurities**
- 4. Polishing:** Further purification steps → polishing chromatography, ultrafiltration, diafiltration to remove any remaining impurities & to **concentrate antibody product**
- 5. Formulation:** Purified antibody is formulated into final product (include buffer exchange, sterile filtration, & addition of stabilizers)
- 6. Viral inactivation and removal:** Ensure the safety antibody product by **inactivating & removing any potential viral contaminants**
- 7. Quality control:** Final product tested to ensure it meets desired purity, potency, & safety specifications

Common Methods in the Industry



1. Protein A chromatography

- *Initial purification*: antibody binds specifically to protein A resin & is eluted with low pH buffer

2. Ion exchange chromatography

- *Polishing*: negatively charged contaminants are separated from positively charged antibody by an ion exchange resin

3. Size exclusion chromatography

- *Final polishing*: large aggregates & small molecular weight contaminants are separated from the antibody based on size

4. Ultrafiltration and diafiltration

- Used for concentration & buffer exchange antibody product

5. Viral inactivation and removal

- Ensure safety by inactivating/removing viral contaminants (pasteurization & nanofiltration)

Issues with Downstream Processing

1. **Contamination byproducts**
2. **High cost** – equipment, raw material
3. **Time consuming** – can take weeks or months
4. **Scalability** – maintain quality & consistency from lab to industry scale
5. **Cell culture medium complexity** - contains variety of proteins/impurities, making it challenging to separate/purify antibody

Proposal for Issues/Impact

Hypothesis

1. Engineering optimal cell cultures

- Spend huge portions on engineering cell cultures/strains → contain less proteins/impurities & yield less contaminants

2. Reduce time consumption

- Implement recycling of biomass back to fermenters after clarification → speeds process
- Innovative purification technologies → continuous chromatography or membrane-based separations → increase efficiency & scalability of downstream processing

3. Implement machine learning

- Use to predict purity and yield based on cell culture conditions, purification steps, & quality control
- Use to automate quality control → identifying impurities in final product → reduce time/cost of testing
- Use to trouble shoot → analyze past batches & identify issues or trends → prevent future issues

Proposal Aims Implementation



1. Engineering optimal cell cultures

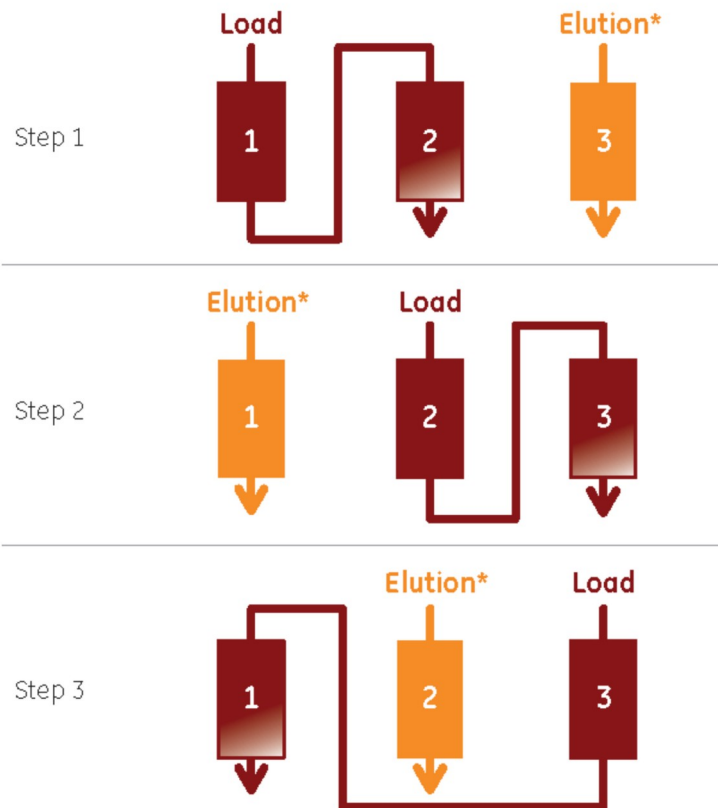
- Establishing a research area & facility → choose best antibody expression
- Develop cell lines optimal for production

2. Reduce time consumption

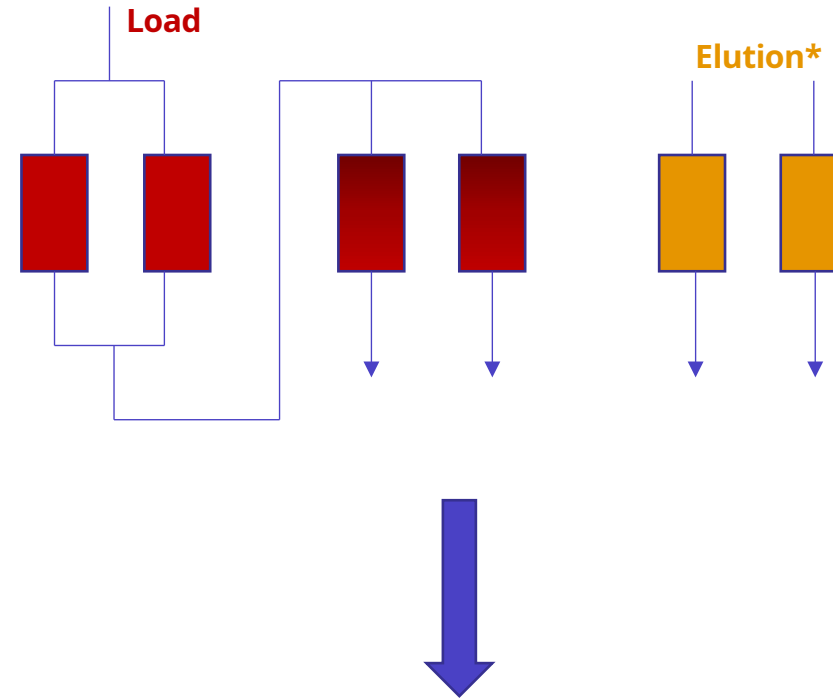
- Recycling biomass → check cell line viability by freeze drying aliquots → run tests & check % viability → analyze growth of culture in lab
- Continuous processes → using multiple equipment in parallel & series that switch from loading/non-loading in a continuous manner (opposed to traditional batch route)

Proposal Aims Implementation

Before



After



Proposal Aims Implementation

3. Implement machine learning

- Define objectives and scope
- Gather and preprocess data
- Identify relevant features/variables that can serve as input for the machine learning model.
 - These could include process parameters (e.g., temperature, pH, flow rate), equipment settings, raw material attributes
- Train model with algorithms based on problem and desired outcome
- Optimize & deploy model into existing software, developing custom APIs, or using a cloud-based platform

Results/Issues

1. Engineering optimal cell cultures

- Can introduce pathogenicity into new strains
- Genetic instability
- New cell cultures can fail at higher scale

2. Reduce time consumption

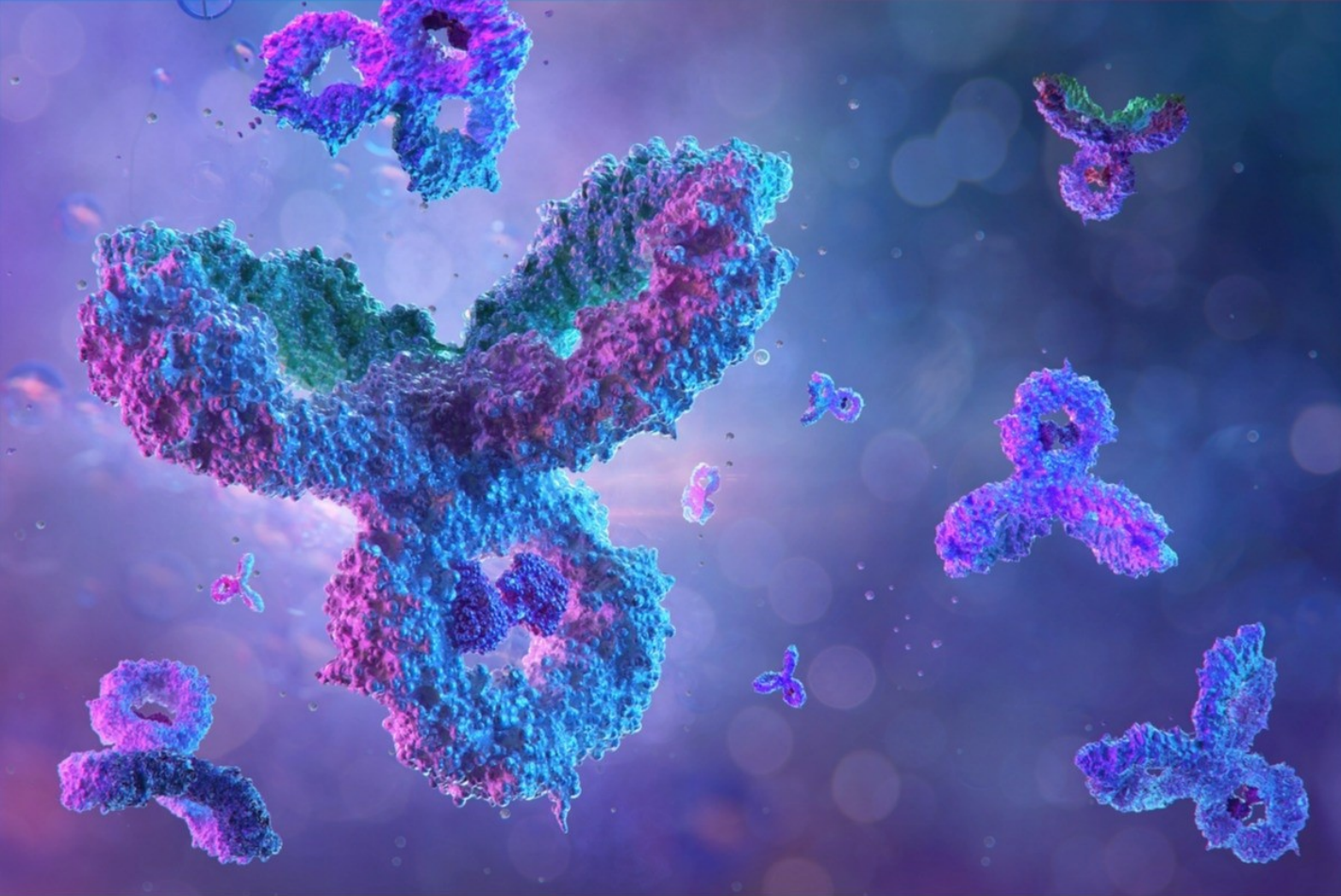
- Equipment (hydrocyclones) used for clarification can shred cells → destroying viability → recycling ineffective

3. Implement machine learning

- Data quality & availability → leads to poor model performance & predictions if data is poor
- Overfitting → capturing noise/random fluctuations instead of underlying patterns
- Integrating machine learning models with existing process control systems, data management platforms, & software tools

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Thank you
Questions?